

MK DATA

Enterprise Fintech & VTU Platform — Client Handover Guide

Official Production Handover Deliverable | Lead Developer: 08034910470

1. Executive Summary & Architecture

MK DATA is an enterprise-grade fintech and VTU (Virtual Top-Up) solution built on Next.js 16 App Router, React 19, Neon Serverless PostgreSQL, Prisma ORM, and TailwindCSS. The platform supports instant mobile data bundling, airtime recharging, electricity token vending, cable TV subscriptions, and WAEC/NECO/JAMB exam PIN purchases. Automated wallet funding is achieved via dedicated BillStack reserved virtual bank accounts with HMAC webhook validation.

2. Technology Stack & Frameworks

Component Layer	Technology / Framework	Operational Role
Frontend & API	Next.js 16, React 19, TypeScript	Web portal, PWA, order checkout & API endpoints
Database	PostgreSQL (Neon Serverless) & Prisma 6	ACID transactions, indexed catalogs & ledgers
Payment Gateway	BillStack Reserved Accounts	Automated bank transfer wallet funding
VTU Telecom APIs	Alrahuz Data, AmySub, Saiful, SMEPlug	Multi-network data & utility dispatch engine
Push Notifications	Firebase Admin SDK (FCM Multicast)	Device push notifications & broadcast notices

3. Third-Party Account Transfer Matrix

Provider	Architecture Role	Required Transfer & Verification Action
Vercel Hosting	Web & API Hosting	Transfer Project ownership; verify custom domain SSL.
Neon PostgreSQL	Production Database	Transfer project to client organization; verify query execution.
BillStack	Payment Gateway	Update KYC, settlement bank, and Webhook URL to live domain.
Alrahuz Data	Primary Telecom API	Transfer portal account, fund float balance, and set API token.
Firebase (GCP)	Push Notifications	Grant Owner role in Firebase Console; generate fresh JSON key.

4. Operational Troubleshooting Matrix

Symptom	Probable Cause	Resolution Procedure
Bank transfer not credited	Webhook URL error or secret mismatch	Verify BILLSTACK_SECRET_KEY in Vercel; check /admin/webhooks log.
Data vending failed	Vendor float balance exhausted	Top up Alrahuz vendor balance via bank transfer immediately.
Push broadcast fails	Firebase service JSON invalid	Re-generate private key from Google Cloud Console & update Vercel.
Admin login rejected	PIN mismatch or unseeded user	Run 'npm run seed:admin-services' with MK_ADMIN_PIN='000000'.

■ **Lead Developer Contact & Technical Handover Support**

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- **Handover Assistance:** Direct Architecture, Database & Environment Walkthroughs